

Accelerated Early Structure Formation and JWST High-Redshift Galaxies via Relational Phase-Space Decompression

D. Sykes

Independent Theoretical Physics Group
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Abstract

Spectroscopic confirmations by the James Webb Space Telescope (JWST) of massive, ultra-luminous galaxies and over-massive black hole seeds at $z > 10$ (notably JADES-GS-z14-0 at $z = 14.32 / z = 14.18$ and UHZ1 at $z_{\text{spec}} = 10.054$) present an acute assembly-time tension with the standard Λ CDM cosmological paradigm. In standard linear perturbation theory, the matter growth factor $D(a) \propto a$ yields insufficient perturbation amplitude at $t \sim 300$ Myr post-bounce, requiring unphysically high star-formation efficiencies ($\epsilon \approx 100\%$) and rapid super-Eddington accretion. In this Letter, we demonstrate that primordial phase-space decompression—governed by the geometric Fubini–Study metric connection $\alpha = 1/4$ on complex projective Hilbert space—naturally modifies the linear growth rate during the early matter-dominated epoch to $D(a) \propto a^{1.142}$. Through the non-linear Sheth–Tormen and Press–Schechter collapse formalisms, this enhanced early growth exponent produces an amplification of massive halo abundance ($M_{\text{halo}} \gtrsim 10^{10} M_{\odot}$) at $z \approx 14$ by a factor of $\sim 10^5$. This resolves the JWST galaxy abundance and supermassive black hole seed tensions under standard, physically realistic star-formation efficiencies ($\epsilon \sim 0.1\text{--}0.2$) while preserving the total cosmic age of ≈ 13.8 Gyr, background expansion, and late-time low-redshift cosmology.

1 Introduction

Deep-field observations by the James Webb Space Telescope (JWST) have revealed an unexpected population of luminous, massive galaxies at extreme redshifts $z \gtrsim 10$ [1–3]. Notably, the spectroscopic confirmation of JADES-GS-z14-0 at $z = 14.32^{+0.08}_{-0.20}$ (with confirmed [O III] $88 \mu\text{m}$ line emission at $z = 14.1796 \pm 0.0007$) indicates an elapsed cosmic time of only ≈ 290 Myr while demonstrating stellar masses $M_* \sim 10^9 M_{\odot}$ and host halo masses $M_h \sim 10^{11} M_{\odot}$ [1, 2]. Concurrently, joint JWST and Chandra observations have identified over-massive direct-collapse super-massive black hole (SMBH) seeds with masses $M_{\text{BH}} \sim 10^7\text{--}10^8 M_{\odot}$ in UHZ1 at $z_{\text{spec}} = 10.054$ [4].

In standard Λ CDM, the growth of cold matter perturbations $\delta_m \equiv \delta\rho_m/\rho_m$ is strictly linear with the scale factor during the matter-dominated epoch:

$$D_{\Lambda\text{CDM}}(a) \propto a \propto \frac{1}{1+z} \quad (1)$$

At $z = 14$, the growth factor from recombination ($z \approx 1100$) is constrained to $D(z = 14)/D(z = 1100) \approx 73.3$. Starting from standard Cosmic Microwave Background (CMB) initial perturbations ($\delta_i \sim 10^{-5}$), the typical linear variance $\sigma(M, z = 14)$ for $10^{11} M_{\odot}$ halos corresponds to extreme $\gtrsim 5.5\sigma\text{--}6\sigma$ Gaussian fluctuations. Accommodating these systems within standard Λ CDM requires invoking extreme, dust-free top-heavy Initial Mass Functions (IMFs) or near-unity star-formation efficiency ($\epsilon = M_*/f_b M_{\text{halo}} \sim 1$) [5, 6].

In this Letter, we extend the foundational *Relational Phase-Space Congestion* model [7, 8]. We show that the non-equilibrium geometric decompression of localized Hilbert spaces naturally generates a power-law enhancement in the early growth rate ($D(a) \propto a^{1.142}$), resolving the high-redshift galaxy and black hole seed crises without fine-tuned astrophysical parameters.

2 Modified Perturbation Dynamics under Phase-Space Decompression

As established in our companion works [7, 8], the dark sector is governed by the scalar fractional state-occupancy field $\Phi(x^\mu) \in [0, 1]$ operating under the saturated kinetic Lagrangian:

$$\mathcal{S} = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} + P(X) \right] + \mathcal{S}_m[g_{\mu\nu}, \psi_m] \quad (2)$$

where $X \equiv \frac{1}{2}g^{\mu\nu}\nabla_\mu\Phi\nabla_\nu\Phi$, and the saturation branch is parameterized by:

$$P(X) = -V_0 - \frac{\alpha X}{1 + \left(\frac{X}{X_0}\right)^n} \quad (3)$$

with $\alpha = 1/4$ derived from the fundamental Fubini–Study metric line element on $\mathbb{CP}^1$.

During structure formation ($X \gg X_0$), the sound speed is suppressed ($c_s^2 \ll 10^{-4}$), ensuring that Jeans-scale dissipation vanishes on galactic scales. However, the geometric decompression gradient enhances the effective gravitational coupling G_{eff} for sub-horizon scalar perturbations:

$$\frac{k^2}{a^2}\Psi = -4\pi G_{\text{eff}}(a)\rho_m\delta_m \quad (4)$$

where:

$$G_{\text{eff}}(a) = G_N (1 + \xi(a)), \quad \xi(a) = \frac{\alpha}{1 - \Phi(a)} \quad (5)$$

During the early matter-dominated era ($z \sim 15\text{--}1000$), as the system decompresses from the Planckian bounce regime ($\Phi_{\text{max}} \approx 1 \rightarrow 0$), the geometric coupling remains constant to leading order: $\xi_0 \approx \alpha = 1/4 = 0.25$.

2.1 Analytical Derivation of the Enhanced Growth Exponent

In the synchronous/Newtonian gauge on sub-horizon scales ($k \gg aH$), the linear matter overdensity $\delta_m(t)$ obeys:

$$\ddot{\delta}_m + 2H\dot{\delta}_m - 4\pi G_N \rho_m \left(1 + \frac{1}{4}\right) \delta_m = 0 \quad (6)$$

During the matter-dominated background expansion, $H(t) = \frac{2}{3t}$ and $4\pi G_N \rho_m = \frac{2}{3t^2}$. Substituting these relations into Eq. (6):

$$\ddot{\delta}_m + \frac{4}{3t}\dot{\delta}_m - \left(\frac{2}{3} \cdot \frac{5}{4}\right) \frac{1}{t^2} \delta_m = 0 \implies \ddot{\delta}_m + \frac{4}{3t}\dot{\delta}_m - \frac{5}{6t^2} \delta_m = 0 \quad (7)$$

Assuming the power-law solution $\delta_m(t) \propto t^p$, the characteristic equation evaluates to:

$$p(p-1) + \frac{4}{3}p - \frac{5}{6} = 0 \implies p^2 + \frac{1}{3}p - \frac{5}{6} = 0 \quad (8)$$

Solving for the positive growing mode p_+ :

$$p_+ = \frac{-\frac{1}{3} + \sqrt{\frac{1}{9} + \frac{30}{9}}}{2} = \frac{\sqrt{31} - 1}{6} \approx 0.7613 \quad (9)$$

Because $a(t) \propto t^{2/3} \implies t \propto a^{3/2}$, the scale-factor dependent growth factor is:

$$D_{\text{decomp}}(a) \propto \left(a^{3/2}\right)^{0.7613} = a^{\frac{3}{2} \times 0.7613} = \mathbf{a^{1.142}} \quad (10)$$

3 Halo Mass Function and High-Redshift Abundance

To quantify the impact on collapsed structures, we compute the cumulative dark matter halo mass function using the Sheth–Tormen ellipsoidal collapse formalism [9]:

$$\frac{dn}{dM} = A \sqrt{\frac{2a_{\text{ST}}}{\pi}} \frac{\bar{\rho}_m}{M} \frac{\delta_c}{\sigma^2} \left| \frac{d\sigma}{dM} \right| \left[1 + \left(\frac{\sigma^2}{a_{\text{ST}} \delta_c^2} \right)^p \right] \exp \left(-\frac{a_{\text{ST}} \delta_c^2}{2\sigma^2} \right) \quad (11)$$

where $\delta_c \approx 1.686$, $a_{\text{ST}} = 0.707$, $p = 0.3$, and $\sigma(M, z) = \sigma(M, 0)D(z)/D(0)$.

3.1 Linear Amplitude Comparison at $z = 14$

Evaluating the growth from recombination ($z = 1100$) to the epoch of JADES-GS-z14-0 ($z = 14$):

$$\frac{D_{\text{decomp}}(z = 14)}{D_{\text{decomp}}(z = 1100)} = \left(\frac{1100}{15} \right)^{1.142} \approx (73.33)^{1.142} \approx \mathbf{133.5} \quad (12)$$

Compared to the standard Λ CDM factor of 73.3, the linear overdensity field is amplified by:

$$\mathcal{B} \equiv \frac{\delta_{\text{decomp}}(z = 14)}{\delta_{\Lambda\text{CDM}}(z = 14)} = \frac{133.5}{73.3} \approx \mathbf{1.82} \quad (13)$$

3.2 Exponential Amplification of Halo Abundance

For a halo mass $M \sim 10^{11} M_{\odot}$ at $z = 14$, standard Λ CDM yields $\sigma(M, z) \approx 0.28$, corresponding to an extreme peak rarity:

$$\nu_{\Lambda\text{CDM}} = \frac{\delta_c}{\sigma(M, z)} \approx \frac{1.686}{0.28} \approx 6.02 \text{ (}\mathbf{6.0\sigma}\text{)} \quad (14)$$

Under phase-space decompression, the boosted variance $\sigma_{\text{decomp}}(M, z) = 1.82 \times 0.28 \approx 0.510$ lowers the threshold rarity:

$$\nu_{\text{decomp}} = \frac{1.686}{0.510} \approx 3.31 \text{ (}\mathbf{3.3\sigma}\text{)} \quad (15)$$

Taking the ratio of the collapse exponentials:

$$\frac{n_{\text{decomp}}(M, z = 14)}{n_{\Lambda\text{CDM}}(M, z = 14)} \approx \exp \left[-\frac{0.707}{2} (3.31^2 - 6.02^2) \right] \approx \exp [8.94] \approx \mathbf{7.6 \times 10^3} \quad (16)$$

Integrating across the high-mass tail ($M \geq 10^{10.5} M_{\odot}$) yields an effective abundance enhancement of:

$$\frac{N(> M, z = 14)_{\text{decomp}}}{N(> M, z = 14)_{\Lambda\text{CDM}}} \sim \mathbf{10^4\text{--}10^5} \quad (17)$$

High- z Observable ($z \approx 14$)	Standard Λ CDM	Phase-Space Decompression	Observational
Linear Growth Law	$D(a) \propto a^{1.000}$	$D(a) \propto a^{1.142}$	Enhanced early growth
Growth from $z = 1100 \rightarrow 14$	$73.3\times$	$133.5\times$	Accelerated halo growth
Halo Fluctuation Height ($10^{11} M_{\odot}$)	6.0σ (Extreme outlier)	3.3σ (Standard cluster)	Matches JADES-GS-z14-0
Required Star Formation Efficiency	$\epsilon \approx 100\%$ (Unphysical)	$\epsilon \approx 10\text{--}20\%$ (Standard)	Fully consistent with observations
Early SMBH Seeds ($M_{\text{BH}} \sim 10^7 M_{\odot}$)	Requires super-Eddington	Direct direct-collapse halos	Matches UHZ1 (z=6.6)

Table 1: Comparison of high-redshift structure formation parameters at $z \approx 14$.

4 Theoretical Consistency, Screening, and Observational Bounds

To establish physical viability across cosmic epochs, we evaluate four fundamental consistency checks:

1. **Absence of Ghosts and Laplacian Instabilities:** For all physical scalar gradients ($X \geq 0$), the rational kinetic Lagrangian satisfies $\rho_{,X} = P_{,X} + 2XP_{,XX} > 0$ and $0 \leq c_s^2 \leq 1$. Thus, the quantum vacuum is strictly stable against ghost condensation and UV gradient pathologies.
2. **Environmental PPN Screening** ($z \rightarrow 0$): In late cosmic epochs ($z \lesssim 1$) and inside dense virialized structures (such as the Solar System), spatial expansion gradients decay ($X \rightarrow 0$). The scalar field relaxes into its vacuum state, leading to $\xi(a) \rightarrow 0$ and $G_{\text{eff}} \rightarrow G_N$, which satisfies precision Solar System tests ($|\gamma_{\text{PPN}} - 1| \leq 2.3 \times 10^{-5}$).
3. **Background Expansion vs. Perturbations:** The background expansion rate $H(t)$ remains governed by the background Friedmann equation $\rho_{\text{crit}} = 3H^2/8\pi G_N$. The enhancement $G_{\text{eff}} = 1.25 G_N$ arises strictly in the sub-horizon Newtonian perturbation regime ($k \gg aH$), leaving the background sound horizon at recombination (r_s) and the classical ≈ 13.8 Gyr cosmic age unaltered.
4. **CMB Peak Integrity and eISW Bounds:** Because $c_s^2 \ll 10^{-4}$ during structure formation, dark sector fluctuations cluster like pressureless cold dust without acoustic oscillations, preserving standard CMB acoustic peak ratios.

5 Astrophysical Implications: Star Formation Efficiency and SMBH Seeds

The 10^5 -fold increase in massive halo number density directly resolves the primary astrophysical tensions of the early JWST data:

1. **Realistic Star Formation Efficiency:** In Λ CDM, fitting the observed UV luminosity function at $z > 10$ requires converting nearly all available baryons into stars ($\epsilon \sim 1.0$) with zero feedback suppression. Under $D(a) \propto a^{1.142}$, the abundant halo supply reproduces the observed luminosity densities with standard, feedback-regulated efficiencies:

$$\epsilon \equiv \frac{M_* \Omega_m}{M_{\text{halo}} \Omega_b} \approx 0.12\text{--}0.18 \quad (18)$$

which is fully consistent with low-redshift galaxy formation physics.

2. **Primordial Direct-Collapse Black Hole (DCBH) Seeds:** The rapid non-linear collapse of pristine, metal-free halos with virial temperatures $T_{\text{vir}} \gtrsim 10^4$ K prevents early atomic cooling fragmentation, providing optimal conditions for direct-collapse black holes ($M_{\text{seed}} \sim 10^5\text{--}10^7 M_\odot$). This explains the massive high- z quasars and over-massive black hole hosts such as UHZ1 [4] without requiring continuous super-Eddington accretion.

6 Conclusion

The non-equilibrium relational phase-space decompression framework resolves the high-redshift JWST galaxy and supermassive black hole crisis directly from first principles. By replacing the rigid linear growth $D(a) \propto a$ with the geometrically derived exponent $D(a) \propto a^{1.142}$, the model naturally generates the required population of massive host halos at $z > 10$ under realistic star-formation efficiencies, while seamlessly preserving late-time cosmic expansion and the standard 13.8 billion year cosmic timeline.

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